Exercise 5 :

a)

Let $t = 4k + r$, $r \in \{1, 2, 3, 4\}$ and $k = \left\lfloor \frac{t-1}{4} \right\rfloor$

$$Y_t = \sum_{i=0}^{k} e_{t-4i}$$

so $\quad Var(Y_t) = \sum_{j=0}^{k} Var(e_{t-4j}) = (k+1)\sigma_e^2$

$$\boxed{\gamma_0(t) = \left(\frac{\lfloor t-1 \rfloor}{4} + 1\right)\sigma_e^2}$$

b)

let $m \in \mathbb{N}$

$$\rho(t, t-4m) = \frac{(k-m+1)\sigma_e^2}{\sqrt{(k+1)\sigma_e^2}\sqrt{(k-m+1)\sigma_e^2}}$$

$$\boxed{\rho(t, t-4m) = \sqrt{\frac{k-m+1}{k+1}}}$$

c)

$Y_t - Y_{t-4} = e_t \iff (1 - B^4) Y_t = e_t$ so $S = 4$ and $D = 1$

Conclusion : $ARIMA(0,0,0) \times (0,1,0)$